

# Elementary Mixing Bounds and Experiments for the Representation Technique in Subset Sum

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**Abstract**—Subset sum is solved on every input in about  $2^{n/2}$  steps, and nobody knows whether  $2^{n/3}$  (or anything below  $2^{n/2}$ ) is possible. The representation technique does much better on random inputs, about  $2^{0.291n}$ , and the missing ingredient for general inputs is usually called *mixing*: the many representations of a solution must fall into many residue classes modulo a random prime. This note does not give a faster worst-case algorithm. It gives two elementary facts about mixing and one experiment. First, if the weight- $w$  sums take few residues modulo a prime  $p$ , then the input values themselves are concentrated modulo  $p$ . Second, averaged over a window of primes, the collision energy of the representations is at most  $E_\infty + Y^2 \kappa / \pi$ , which is  $\mathcal{O}(Y \log Y)$  when few exact collisions occur, so a random prime mixes to within a polylogarithmic factor on every input. We then run the complete pipeline—modular filter, balanced sub-solver, disjointness match—on adversarial families. Low mixing appears only on arithmetic structure, which is easy; the hard large-value and generalized-progression families mix at 0.85–0.95 and are solved at the sub-solver cost  $2^{(n/2)_{(n/8)}} = 2^{0.4057n}$ . We conclude that for the coefficient set  $\{0, 1\}$  the obstruction is more likely the balanced sub-solver than mixing, and we state the resulting open problem. The two lemmas are machine-checked in Lean.

**Index Terms**—subset sum, representation technique, mixing, meet-in-the-middle, worst-case analysis

## I. INTRODUCTION

Suppose you have  $n$  integers and a target. Can you pick some of them so that the picked numbers add up exactly to the target? This is the *subset sum* problem. For the numbers 3, 5, 8, 13 and target 16 the answer is yes, in two ways:  $3 + 13$  and  $3 + 5 + 8$ .

Written as an equation, given integers  $a_1, \dots, a_n$  and  $b$ , we ask whether  $\exists x \in \{0, 1\}^n : \sum_i a_i x_i = b$ .

### A. Why it matters

Subset sum is one of the standard NP-complete problems. Trying every subset costs  $2^n$  steps; meet-in-the-middle of Horowitz and Sahni [1] brings this to about  $2^{n/2}$ , and that is still the best bound that holds on every input, fifty years later. Schroeppe and Shamir [2] kept the  $2^{n/2}$  time with  $2^{n/4}$  memory, and Chen, Jin, Randolph and Servedio [5] removed a polynomial factor, but the exponent stayed  $1/2$ . Whether  $2^{(1/2-\varepsilon)n}$  is possible is open.

### B. What is known

On *random* inputs the representation technique of Howgrave-Graham and Joux [3], refined by Becker, Coron and Joux [4], runs in about  $2^{0.291n}$  steps. Its speed comes from splitting the *answer* rather than the positions: a solution has enormously many two-piece *representations*, and after filtering

them with a random prime only a small subfamily remains to be searched. The analysis assumes the representations are spread across the residue classes—they *mix*. Randolph and Węgrzycki [6] made the technique work on every input for many coefficient sets, using a “mixing dichotomy”; the plain subset sum case  $\{0, 1\}$  resisted. Austrin, Kaski, Koivisto and Nederlof [7] showed that instances with a very small or very large maximum bin size are also easy, leaving a hard middle band.

### C. What this note does

- It sets up the mixing condition and defines the coverage and the collision energy (Section II), with a worked example.
- It proves two elementary lemmas: few hit residues force the input values to be concentrated modulo the prime, and one residue class can be stripped and the instance recursed (Section III). Both are machine-checked in Lean.
- It proves that the average collision energy over a window of primes is  $\mathcal{O}(Y \log Y)$  when few exact collisions occur, so a random prime mixes to within a polylogarithmic factor on every input (Section IV).
- It runs the full pipeline on adversarial families and finds that low mixing coincides with easy arithmetic structure, while hard families mix and run at the sub-solver cost (Section V).

Section VI states what the experiments suggest the real obstruction is. We do not claim a faster worst-case algorithm.

### D. A worked example of mixing

Let the support be  $A = \{3, 5, 8, 13\}$  and let  $w = 2$ , so the representations are the 2-subsets. Their sums are 8, 11, 16, 13, 18, 21. Modulo  $p = 7$  these are 1, 4, 2, 6, 4, 0, so  $s_p = 5$  classes are hit and the coverage is  $5/7 \approx 0.71$ . Modulo  $p = 5$  they are 3, 1, 1, 3, 3, 1, so only  $s_p = 2$  classes are hit and the coverage is 0.4. A random prime rarely looks like  $p = 5$ ; the point of this note is to say how rarely, and what poor coverage implies.

## II. PRELIMINARIES AND THE MIXING CONDITION

Let  $A = \{a_1, \dots, a_m\} \subseteq \mathbb{Z} \setminus \{0\}$  and let  $w$  with  $1 \leq w \leq m - 1$ . In the balanced regime  $m = n/2$  is the support of a solution and  $w = n/4$ . Write  $[m] = \{1, \dots, m\}$ , and

$$\mathcal{Y} = \{y \subseteq [m] : |y| = w\}, \quad \Sigma(y) = \sum_{i \in y} a_i, \quad Y = \binom{m}{w},$$

so  $\mathcal{Y}$  is the set of *representations*. For a prime  $p$  let

$$c_r = \#\{y \in \mathcal{Y} : \Sigma(y) \equiv r \pmod{p}\}, \quad E_p = \sum_r c_r^2,$$

the *collision energy*, and let  $s_p = \#\{r : c_r > 0\}$  be the number of residue classes hit. The filter keeps the sums modulo  $p$ ; a representation survives in a random class with probability  $s_p/p$ , so the technique needs  $s_p$  close to  $p$ . We call  $s_p/p$  the *coverage*. Since  $\sum_r c_r = Y$  and there are  $s_p$  nonzero terms, Cauchy–Schwarz gives

$$E_p \geq Y^2/s_p. \quad (1)$$

The *exact* collision energy is  $E_\infty = \sum_v d_v^2$ , where  $d_v = \#\{y : \Sigma(y) = v\}$ .

### III. MIXING FORCES CONCENTRATION

The first lemma says that poor mixing is a statement about the values, not merely about their weight- $w$  sums. All proofs are in Appendix A.

**Lemma 1** (Concentration).  $|(A \bmod p) - (A \bmod p)| \leq s_p^2$ , and hence  $|\{a_i \bmod p\}| \leq s_p^2 + 1$ .

Poor mixing therefore forces additive structure. Structure is not the same as a small sumset:  $A = \{p2^0, \dots, p2^{m-1}\}$  has  $s_p = 1$  yet  $|S(A)| = 2^m$ ; it is structured, and easy. The single-residue case is exact.

**Lemma 2** (Contraction). *If  $p \mid a_i - \rho$  for all  $i$ , then every representation sum satisfies  $\Sigma(y) \equiv w\rho \pmod{p}$ . Writing  $a_i = \rho + pb_i$ , every subset sum is  $|y|\rho + p\Sigma_b(y)$ , so  $p$  can be stripped from the values and the instance recursed.*

For larger  $s_p$ , the analogue is that  $A \bmod p$  lies in a short generalized arithmetic progression (Kneser’s theorem); we do not pursue that step here.

### IV. AVERAGE COLLISION ENERGY OVER PRIMES

The next result holds for *every* input, on average over primes.

**Lemma 3** (Average energy). *Let  $B = \max_i |a_i|$ , let  $\kappa = \lceil \log(2wB)/\log P \rceil$ , and let  $\pi = \#\{p \text{ prime} : P \leq p \leq 2P\}$ . Then*

$$\frac{1}{\pi} \sum_{p \in [P, 2P]} E_p \leq E_\infty + \frac{Y^2 \kappa}{\pi}.$$

Take  $P \approx Y$ , so  $\pi \approx Y/\ln Y$ , and  $B \leq 2^{\beta n}$  with  $m, w = \Theta(n)$ ; then  $\kappa \leq 2\beta + o(1)$ . If few exact collisions occur, i.e.  $E_\infty \leq Y \text{polylog}(Y)$ , Lemma 3 gives an average of  $\mathcal{O}(Y \log Y)$ . Markov’s inequality controls the tail: for all but an  $\mathcal{O}(1/\log Y)$  fraction of primes,  $E_p = \mathcal{O}(Y \log^3 Y)$ , so by (1)

$$s_p \geq \frac{Y^2}{E_p} = \frac{Y}{\text{polylog}(Y)}. \quad (2)$$

A representation is therefore found after  $\text{polylog}(Y)$  retries, and at least half of the primes mix to coverage  $Y/\text{polylog}$ .

Table I checks the bound directly. The average energy is about  $2Y$ , the birthday value, well below the bound and below

TABLE I  
AVERAGE ENERGY AND COVERAGE OVER EVERY PRIME  $p \in [Y, 2Y]$ , FOR A PLANTED BALANCED SUPPORT WITH SPARSE VALUES AND NO EXACT COLLISIONS.

| $n$ | $Y$ | primes | avg. $E_p$ | bound $Y^2 \kappa / \pi$ | min. coverage |
|-----|-----|--------|------------|--------------------------|---------------|
| 16  | 70  | 15     | 113        | 653                      | 0.371         |
| 20  | 252 | 43     | 422        | 2954                     | 0.325         |
| 22  | 462 | 68     | 778        | 6278                     | 0.270         |

TABLE II  
FULL PIPELINE ON ADVERSARIAL FAMILIES AT  $n = 40$ . COVERAGE IS THE FRACTION OF RESIDUES HIT BY THE PLANTED SOLUTION’S REPRESENTATIONS; WORK IS THE MEAN ENUMERATION, BASELINE  $2^{0.4057n}$ .

| family                  | success | coverage | work            |
|-------------------------|---------|----------|-----------------|
| random $\beta = 1$      | 1.00    | 0.95     | $2^{0.4057n}$   |
| random $\beta = 1.5$    | 1.00    | 0.95     | $2^{0.4057n}$   |
| two-scale $qb_i + r_i$  | 1.00    | 0.94     | $2^{0.4057n}$   |
| rank-2 generalized AP   | 1.00    | 0.91     | $2^{0.4057n}$   |
| $q$ -multiple           | 1.00    | 0.95     | $2^{0.4057n}$   |
| geometric (superincr.)  | 0.17    | 0.94     | $> 2^{0.4057n}$ |
| arithmetic (large step) | 0.33    | 0.003    | $> 2^{0.4057n}$ |

$Y \log Y$ ; and even the *pointwise minimum* coverage over all primes stays at 0.27–0.37, not only the average.

### V. THE FULL PIPELINE ON ADVERSARIAL FAMILIES

Lemma 3 concerns all weight- $w$  sums; the algorithm needs a representation of the actual solution. We therefore ran the complete pipeline: enumerate weight- $n/8$  subsets of each half and match by residue (the balanced sub-solver,  $2^{\binom{n/2}{n/8}} = 2^{0.4057n}$  work per residue), filter by a random prime, and look for a disjoint pair of weight- $n/4$  pieces with exact sum  $b$ . Table II reports, for planted balanced solutions, the fraction solved within 16 residues, the coverage, and the enumeration work.

The hard families—large random values, two-scale weights  $qb_i + r_i$ , rank-2 generalized progressions, near-multiples—all mix at 0.91–0.95 and are solved at the sub-solver cost, up to a small constant. The only low-coverage family is the large-step arithmetic progression (coverage 0.003), which is additively structured and easy. Geometric (superincreasing) instances are high-coverage but the balanced enumeration happens to fail on them, and they are solved greedily in any case. No hard, poorly-mixing family was found.

### VI. WHERE THE BARRIER LIES

Combining the lemmas and the experiments: on every input a random prime mixes to within a  $\text{polylog}$  (Lemma 3), and in the tests the only low-coverage families are structured and easy. If mixing were the obstruction, the pipeline would already beat  $2^{n/2}$  using the balanced sub-solver, since  $0.4057 < 1/2$ . It does not, so for  $\{0, 1\}$  the obstruction must lie elsewhere. The experiments isolate two candidates.

(a) *The balanced sub-solver.* The filtered class is enumerated by matching weight- $n/8$  subsets of two fixed halves, which finds only representations that are weight-balanced across the

split. An adversary can force the solution’s representations off balance, and the recursion that replaces the balanced sub-solver—the four-way decomposition of [3], [4]—is where the published  $0.291n$  bound is won; without it the exponent is stuck at 0.4057.

(b) *Compatibility and exactness.* Two pieces may combine to an invalid coefficient vector (a “pseudo-solution”), and recovering a compatible pair with an exact, not merely modular, sum is a sparse orthogonal-vectors step whose cost must be kept list-linear.

The open problem suggested by the experiments is therefore:

*For every  $\{0, 1\}$  instance, either a solution has a representation that the balanced sub-solver enumerates and the compatibility certificate recovers, or the instance is solved below  $2^{n/2}$  by other means; and the filtered class is enumerable and searchable in  $2^{(1/2-\epsilon)n}$  for a constant  $\epsilon > 0$ .*

*Remark on a recent preprint.* A 2025 preprint [8] claims a guaranteed  $2^{n/2-\epsilon}$  via a “controlled aliasing” that replaces one element by another inside a half’s partial sums and keeps one canonical subset per aliased sum. The mechanism is unsound: in the half  $\{1, 2\}$ , aliasing  $2 \rightarrow 1$  gives the canonical subset  $\{1, 2\}$  a stored sum of 2 although its true sum is 3, so the merge can return a false positive and the subset  $\{2\}$  is lost. The claimed theorem is deferred to a companion paper and is not established.

## VII. CONCLUSION

Mixing in the representation technique obeys an elementary, input-independent bound: on average over primes the collision energy is  $\mathcal{O}(Y \log Y)$ , and pointwise coverage stayed above 0.27 on every input we tested. In the full pipeline the hard, unstructured families mix at 0.9 and run at the balanced sub-solver cost, while low mixing coincides with arithmetic structure and easy inputs. For the coefficient set  $\{0, 1\}$  this points to the sub-solver and the compatibility step, not mixing, as the obstruction. A genuinely hard, poorly-mixing family, if one exists, was not found; finding one, or proving none exists, is the next step.

## APPENDIX A PROOFS

### A. Proof of Lemma 1

Fix  $i, j$ . Since  $1 \leq w \leq m-1$  and  $m \geq 2$ , the set  $[m] \setminus \{i, j\}$  has at least  $m-2 \geq w-1$  elements, so there is  $S \subseteq [m] \setminus \{i, j\}$  with  $|S| = w-1$ . The sets  $S \cup \{i\}$  and  $S \cup \{j\}$  both have size  $w$ , so  $u = a_i + \Sigma(S)$  and  $v = a_j + \Sigma(S)$  are both values of weight- $w$  sums. Their difference is  $u - v = a_i - a_j$ . Letting  $i, j$  vary, every value difference modulo  $p$  is a difference of two hit residues; the hit residues number  $s_p$ , so the difference set has at most  $s_p^2$  elements. Fixing one index as a base gives  $|\{a_i \bmod p\}| \leq s_p^2 + 1$ .  $\square$

### B. Proof of Lemma 2

If  $a_i \equiv \rho \pmod{p}$  for all  $i$ , then  $\Sigma(y) = \sum_{i \in y} a_i \equiv |y|\rho \equiv w\rho \pmod{p}$ . Writing  $a_i = \rho + pb_i$  with  $b_i = (a_i - \rho)/p \in \mathbb{Z}$ , we get  $\Sigma(y) = |y|\rho + p\Sigma_b(y)$ , so the target determines  $p\Sigma_b(y)$  given  $|y|$ .  $\square$

### C. Proof of Lemma 3

For each prime  $p$ ,  $E_p = \sum_r c_r^2 = \#\{(y, y') : \Sigma(y) \equiv \Sigma(y') \pmod{p}\}$ , so  $\sum_p E_p = \sum_{y, y'} n_{y, y'}$  with  $n_{y, y'} = \#\{p \in [P, 2P] : p \mid \Sigma(y) - \Sigma(y')\}$ . If  $\Sigma(y) = \Sigma(y')$  then  $n_{y, y'} = \pi$ ; there are  $E_\infty$  such ordered pairs. Otherwise  $N = \Sigma(y) - \Sigma(y') \neq 0$  and  $|N| \leq 2wB$  because each value is at most  $B$  in absolute value and a weight- $w$  sum has at most  $w$  terms; the number of primes in  $[P, 2P]$  dividing  $N$  is at most  $\lfloor \log |N| / \log P \rfloor \leq \kappa$ . The remaining  $Y^2$  ordered pairs each contribute at most  $\kappa$ . Summing and dividing by  $\pi$  gives the claim.  $\square$

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